Smart Community Health Monitoring and Early Warning System for Water-Borne Diseases in Rural Northeast India

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Abstract: Waterborne diseases continue to pose a serious threat to public health, particularly in rural regions where access to clean drinking water and proper sanitation remains limited. This study proposes a machine learning-based health monitoring and early warning system designed to predict disease risk levels using environmental and case-related data. Multiple classification algorithms, including Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Logistic Regression, are implemented to categorize risk into low, medium, and high levels. The system evaluates model performance using metrics such as accuracy and ROC-AUC. A web-based interface developed using Flask enables real-time prediction and visualization of results. The findings demonstrate that integrating machine learning with real-time monitoring systems can significantly enhance early detection and support informed decision-making in public health management.

I. INTRODUCTION

Waterborne diseases such as diarrhea, cholera, typhoid, and hepatitis A remain major contributors to illness and mortality, particularly in developing regions with inadequate sanitation and limited access to safe drinking water. These diseases are primarily transmitted through contaminated water sources and continue to place a significant burden on healthcare systems.

Traditional methods of disease detection rely heavily on manual data collection and reporting, which are often time-consuming and prone to inaccuracies. As a result, delays in identifying outbreaks can lead to rapid disease spread and increased health risks. With the advancement of artificial intelligence and machine learning techniques, there is an opportunity to develop automated systems capable of analyzing historical data and predicting potential outbreaks more efficiently.

Machine learning models can identify hidden patterns in data and provide early warnings based on trends such as seasonal variations and reported case counts. In this study, we focus on developing an intelligent prediction system tailored for rural regions, where early intervention is critical. The proposed system utilizes seasonal

information and total disease case data to classify risk levels and assist authorities in implementing timely preventive measures.

II. PROPOSED WORK

A. Aim and Objectives

The primary objective of this study is to develop a machine learning-based system capable of predicting the risk levels of waterborne diseases. The system is designed to analyze disease-related data, classify risk into different categories, and support early warning mechanisms.

The specific objectives of the project include:

- Performing preprocessing and analysis of disease datasets to extract meaningful features
- Categorizing disease risk levels into low, medium, and high classes
- Implementing and comparing multiple machine learning algorithms
- Evaluating model performance using metrics such as accuracy and ROC-AUC
- Developing a web-based application for real-time prediction and user interaction

B. Research Design

- The research design of the proposed system consists of multiple stages, including data preparation, model development, evaluation, and deployment.

1. Data Collection and Preprocessing
The dataset used in this study includes recorded cases of major waterborne diseases such as diarrhea, cholera, typhoid, and hepatitis A. A new feature, termed *Total Cases*, is generated by aggregating the number of reported cases across all diseases. Seasonal information is converted into numerical format to enable model processing. The dataset is further cleaned and normalized to improve data quality and consistency.

2.ModelDevelopment
Several machine learning algorithms are implemented to perform classification, including Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Logistic Regression. These models are trained using the processed dataset to learn patterns associated with disease risk levels.

3.PerformanceEvaluation
The effectiveness of the models is assessed using evaluation metrics such as accuracy, precision, recall, and the ROC-AUC curve. These metrics help in identifying the most suitable model for prediction.

4.Deployment
The final model is integrated into a Flask-based web application, allowing users to input relevant parameters such as season and number of cases. The system then generates real-time predictions of disease risk levels.

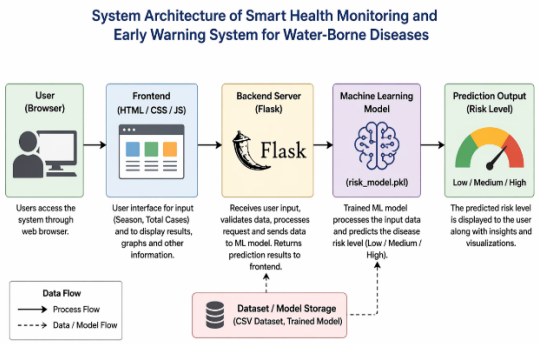

Fig 5.1 System Architecture of Smart Health Monitoring and Early Warning System for Water-Borne Diseases

III. ALGORITHM

Machine learning classification algorithms are used to predict disease risk levels.

- Random Forest: Ensemble model combining multiple decision trees
- SVM: Separates data using hyperplanes
- KNN: Classifies based on nearest neighbours
- Logistic Regression: Probability-based classification

$$P(y=1) = 1 / (1 + e^{(-z)})$$

$P(y=1) \rightarrow$ Probability of high disease risk

$z \rightarrow$ Weighted sum of input features

This formula is used to calculate the probability of disease risk.

Among these, Random Forest provided the best performance due to its ability to handle non-linear data and reduce overfitting.

IV. DATASET

The dataset includes:

- Season (encoded as numbers)
- Number of disease cases

A new column called Total Cases is created by adding all disease cases.

Based on total cases:

- Low Risk → less than 50
- Medium Risk → 51 to 149
- High Risk → above 150

The dataset is used to train and test the models.

V. METHODOLOGY

The system follows these steps:

1. Data Preprocessing
- Cleaning dataset
- Encoding season values
- Creating total cases
2. Feature Selection
- Inputs: Season, Total Cases
3. Model Training
- Train multiple ML models
4. Prediction
- Model predicts risk level
5. Evaluation
- Accuracy comparison
- ROC-AUC curve

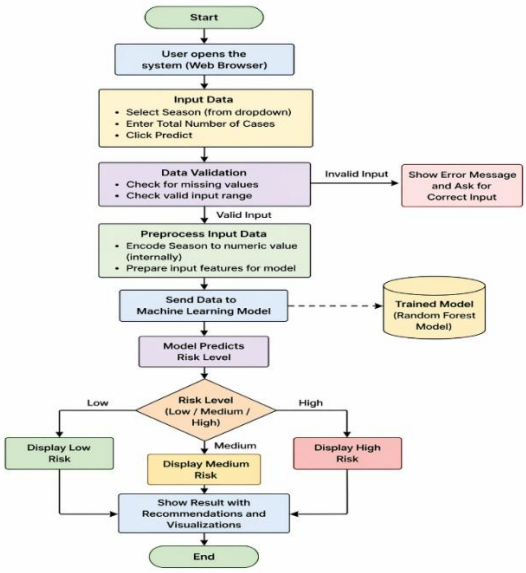

Fig 5.2 System Flow Chart for Water-Borne Disease Prediction System

6. Web Application

- Built using Flask
- User inputs data
- System shows prediction result

Fig 3: Home Page

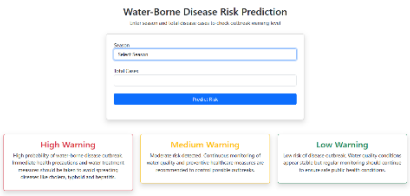

Fig 4: Prediction Page

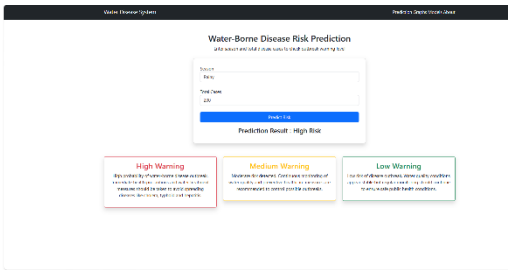

Fig 5: Output Result

VI. RESULTS

The system successfully predicts disease risk levels.

- Random Forest achieved highest accuracy
- ROC-AUC curve shows model performance
- Web app provides real-time prediction

This system helps in early warning and reduces disease spread.

VII. CONCLUSION

This project shows how machine learning can be used to accurately predict the risk of diseases spread by water.

The system is easy to use, quick, and useful in the real world. It helps find problems early, make better choices, and make people's health better.

Improvements that could happen in the future are:

Adding data in real time

Using sensors in the Internet of Things

Getting more data to make things more accuracies.

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